Bures-sur-Yvette, March 9, 1974

Dear Looijenga,

I spoke about your problem with Tits and Don Zagier, and they solved it for me.

1. Coxeter transformations

1.1

Let W be a finite Coxeter group, D its Dynkin diagram, V the space of its natural representation, and $R \subset W$ the set of reflections. I shall call a *root* a unit vector orthogonal to a reflecting hyperplane. If α is a root, write

$$V^+(\alpha) = \{x \mid (x, \alpha) \geq 0\}, \qquad M(\alpha) = \partial V^+(\alpha)$$

for the corresponding wall, and $z_\alpha \in R$ for the reflection with respect to $M(\alpha)$. Assume that D is irreducible and $D \neq A_1$.

1.2

Let c be a Coxeter transformation. The following facts are proved in Bourbaki, *Groupes et algèbres de Lie*, Chapter V, §6, no. 2.

(i) Let h be the order of c, and let $P \subset V$ be the largest subspace of V on which c has only the eigenvalues $\exp(\pm 2\pi i/h)$. Then $\dim(P) = 2$. Orient P so that $c|_P$ is a rotation through angle $2\pi/h$.

(ii) No wall contains P; the walls cut on P a system of reflecting hyperplanes of type $I_2(h)$.

(iii) The walls of V containing the same wall d of P are pairwise orthogonal. Write $s(d)$ for the involution that is the product of the corresponding reflections.

(iv) Let C_0 be a chamber of P, let C be the unique chamber of V containing it, let d_1^+ and d_2^+ be the half-lines bounding C_0, taken in the positive direction, and let d_1, d_2 be the corresponding lines. Then the walls of C are those containing d_1 or d_2, and

$$c = s(d_2)s(d_1).$$

Moreover, $s(d_i)$ stabilizes P, its restriction to P is the reflection with respect to d_i, and $c|_P$ is therefore a Coxeter transformation of P.

Let $c \in W$ be a Coxeter element, and let α be a root.

Corollary (1.3). *(i) With the notation of 1.2(iv), the dihedral group W' generated by $s(d_1)$ and $s(d_2)$ is the stabilizer of P. It is also the stabilizer of $\{c, c^{-1}\}$. The group W' acts faithfully on P, and its restriction to P is the Weyl group of P. The centralizer of c is the subgroup $\langle c\rangle$ generated by c.*

(ii) There is a chamber C in which α is a simple root, and an ordering $(\alpha_1, \dots, \alpha_\ell)$ of the simple roots of C, with $\alpha = \alpha_1$, such that

$$c = z_{\alpha_1} \cdots z_{\alpha_\ell}.$$

(iii) For C as in (ii), the line d obtained as the intersection of the walls $M(\alpha_i)$, $i \neq 1$, is the subspace of V fixed by $z_\alpha c$. Thus it does not depend on C. The subgroup W_d of W fixing d is a Coxeter group, generated by the z_{α_i} for $i \neq 1$; its natural representation is on V/d, and $z_\alpha c \in W_d$ is a Coxeter transformation of W_d.

(iv) The vertex of D corresponding to the wall $M(\alpha)$ of C, with C as in (ii), is independent of C. It is denoted by $\bar{\alpha}$.

(v) $\langle c \rangle$ acts freely on the set of roots. Two roots lie in the same orbit if and only if they have the same image in D.

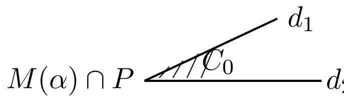
(vi) Let C_0 and C be as in 1.2(iv). For each wall M of C, let $\alpha = \alpha(M)$ be the root such that $M = M(\alpha)$ and such that $V^+(\alpha)$ contains, respectively does not contain, C, according as M passes through d_2, respectively through d_1. Then

$$\{\alpha(M) \mid M \text{ a wall of } C\}$$

is a section of the set of $\langle c \rangle$-orbits, and $\bar{\alpha}(M)$ is the class in D of the wall M of C.

Proof. For (i), W' stabilizes $\{c, c^{-1}\}$, since $s(d_i)cs(d_i)^{-1} = c^{-1}$. Conversely, $\{c, c^{-1}\}$ determines P; hence the stabilizer of $\{c, c^{-1}\}$ stabilizes P. An element $w \in W$ that stabilizes a chamber C_0 of P is the identity, because C_0 is contained in a unique chamber. Since W' is transitive on the chambers of P, it is the full stabilizer of P. Finally, $\langle c \rangle$ is the centralizer of c in W', hence also in W.

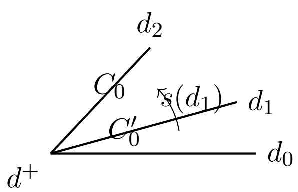
For (ii), take the chamber C_0 of P whose wall is $M(\alpha) \cap P$, such that, with the notation of 1.2(iv), $M(\alpha) \cap P = d_2$ and $V^+(\alpha) \supset C_0$. Since $c = s(d_2)s(d_1)$, this choice works.



For (iii), $z_\alpha c = z_{\alpha_2} \cdots z_{\alpha_\ell}$; the first assertion follows easily. That $z_\alpha c$ is Coxeter is seen by taking C as in (ii).

For (iv), we check that $\bar{\alpha}$ depends only on α and d. If C' and C'' are two chambers in which α is a simple root, and such that the half-line opposite the wall $M(\alpha)$ is $d^+ = d \cap V^+(\alpha)$, then there exists $w \in W_d$ sending C' to C''; indeed, the chambers containing d^+ correspond bijectively to those of V/d. Since w preserves d, it preserves $M(\alpha)$, and (iv) follows.

Writing $c = s(d_2)s(d_1)$, one sees at once, for M passing through d_2, that $\bar{\alpha}(M)$ is the class of the wall M of C. Let $C_0' = s(d_1)C_0$, so $C' = s(d_1)C$; the walls passing through d_1 are treated in the same way. It is clear that the $\bar{\alpha}(M)$ form a section of the $\langle c \rangle$-orbits in the set of roots, and that the action of $\langle c \rangle$ is free. Since $c^{-1}\alpha$ is carried along in the same way, (v) and (vi) follow.



$\square$

Remark (1.4). *No longer assume that D is irreducible or different from A_1. Assertions (ii), (iii), and (iv) still hold. The second part of (v) holds after replacing $\langle c \rangle$ by the centralizer of c. Write $h(D)$ for the order of this centralizer; then*

$$h\Big(\coprod D_i\Big) = \prod h(D_i).$$

Corollary (1.5). *Let W be a finite Coxeter group and let $c \in W$ be a Coxeter element. Let Ξ_c be the set of sequences of ℓ roots $\alpha_1, \ldots, \alpha_\ell$, and let Ξ'_c be the set of sequences of ℓ reflections $z_1, \ldots, z_\ell$, where $\ell = \mathrm{rank} W$, such that*

$$z_{\alpha_1} \cdots z_{\alpha_\ell} = c, \qquad z_1 \cdots z_\ell = c.$$

The numbers

$$F(D) = \#\Xi_c, \qquad F'(D) = \#\Xi'_c$$

depend only on D, and satisfy $F(D) = 2^\ell F'(D)$. For irreducible D one has

$$F(D) = h(D) \sum_{i \in D} F(D - \{i\}),$$

hence

$$F'(D) = \frac{h(D)}{2} \sum_{i \in D} F'(D - \{i\}).$$

Indeed, by 1.3(ii),

$$F(D) = \sum_{\alpha \text{ root}} F(D - \{\bar{\alpha}\}),$$

and one applies 1.3(v), supplemented by 1.4.

Remark (1.6). *One has $F(\varnothing) = F'(\varnothing) = 1$. If $D = \coprod D_i$ and D_i has rank ℓ_i, while $\ell = \sum \ell_i$, then*

$$F(D) = \frac{\ell!}{\prod \ell_i!} \prod_i F(D_i),$$

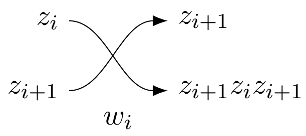
and similarly for F'.

Remark (1.7). *For fixed c, we have defined a map $\alpha \mapsto \bar{\alpha}$ from the set of roots to D. The image $\overline{-\alpha}$ is the transform of $\bar{\alpha}$ by the opposition involution. Thus, if D' denotes the quotient of D by the opposition involution, $\alpha \mapsto \bar{\alpha}$ descends to the quotient and defines a map $z \mapsto \bar{z} : R \to D'$.*

2. Action of the braid group

Let W be a finite Coxeter group, D its Dynkin diagram, D' the quotient of D by the opposition involution, ℓ the rank of W, c a Coxeter transformation, and Ξ'_c as in 1.5. Let B_ℓ be the braid group on ℓ strands, with canonical generators $(w_i)_{1 \le i \le \ell - 1}$. Make it act on Ξ'_c by

$$w_i : (z_1, \ldots, z_\ell) \longmapsto (z_1, \ldots, z_{i-1}, z_{i+1}, z_{i+1} z_i z_{i+1}, z_{i+2}, \ldots, z_\ell).$$



Theorem (2.1). *B_ℓ acts transitively on Ξ'_c.*

Proof. Proceed by induction on ℓ. If D is a disjoint union of diagrams D_i, and if the theorem is true for the D_i, then it is true for D. We are therefore reduced to assuming D irreducible. The cases $D = \varnothing$ and $D = A_1$ are trivial.

Lemma (2.1.1). *Let $\sigma' = (z'_1, \ldots, z'_\ell)$ and $\sigma'' = (z''_1, \ldots, z''_\ell)$ be in Ξ'_c. If $z'_1 = z''_1$, then σ' and σ'' lie in the same B_ℓ-orbit.*

Let d be the subspace of V fixed by $z_1'c$. Apply the induction hypothesis to W_d and to $\Xi'_{z_1'c}$, using 1.3(iii).

Say that two roots α and β are equivalent if there exist $\sigma' = (z_1', \ldots, z_\ell')$ and $\sigma'' = (z_1'', \ldots, z_\ell'')$ in Ξ'_c, conjugate under B_ℓ, such that $z_1' = z_\alpha$ and $z_1'' = z_\beta$. It remains to prove that any two roots are equivalent.

(a) α and $c^{-1}\alpha$ are equivalent. Take $\sigma = (z_\alpha, z_2, \ldots, z_\ell)$ in Ξ'_c; then

$$w_1 \cdots w_{\ell-2} w_{\ell-1} w_{\ell-2} \cdots w_1(\sigma) = (c^{-1} z_\alpha c, \ldots).$$

(b) α and $-\alpha$ are equivalent, since $z_\alpha = z_{-\alpha}$.

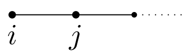
By 1.3(v) and 1.7, there is therefore an equivalence relation on D' such that $\alpha \sim \beta$ if and only if $\bar{\alpha}$ and $\bar{\beta}$ have equivalent images in D'. Since W acts trivially on D', this equivalence does not depend on the choice of c. Write $\sim$ also for the inverse-image equivalence on D.

(c) If i, j are two unjoined vertices of D, then $i \sim j$. Take $c = z_i z_j \cdots$, a product of fundamental reflections. Then

$$w_1(z_i, z_j, \ldots) = (z_j, z_i, \ldots),$$

and $\bar{z}_i = i$, $\bar{z}_j = j$.

(d) For i, j as below, with i an end vertex, one has $i \sim j$.



Take $c = z_j z_i \cdots$, a product of fundamental reflections. Then

$$w_1(z_j, z_i, \ldots) = (z_i, z_i z_j z_i, \ldots),$$

where $z_i z_j z_i$ is the fundamental reflection for the chamber $z_i C$, and $\bar{z}_j = j$, $\bar{z}_i = i$.

Parts (c) and (d) give the equivalence of all vertices, and hence the theorem. $\square$

3. Calculation of $F'(D)$

Corollary (3.1).

$$F'(A_\ell) = (\ell+1)^{\ell-1}.$$

In this case, you have in fact proved your conjecture directly. On the other hand, by 2.1, Ξ'_c is a single B_ℓ-orbit, and one applies your computation of the index.

This is also equivalent to the formula you give for trees: for a product of ℓ transpositions in a set with $\ell+1$ elements to be an $(\ell+1)$-cycle, it is necessary and sufficient that the graph obtained by joining the points appearing in these transpositions be a tree. Since there are $\ell!$ $(\ell+1)$-cycles and $\ell!$ total orderings of a given tree, one has

$$F'(A_\ell) = \#\{\textit{trees with support a given set of } (\ell+1) \textit{ elements}\}.$$

3.2

Corollary 1.5 gives

$$F'(A_\ell) = \frac{\ell+1}{2} \sum_{\substack{i+j=\ell-1 \\ i,j\geq 0}} \frac{(\ell-1)!}{i!j!} F'(A_i)F'(A_j),$$

or

$$2\ell \frac{F'(A_\ell)}{\ell!} = (\ell+1) \sum_{i+j=\ell-1} \frac{F'(A_i)}{i!} \frac{F'(A_j)}{j!}.$$

In terms of the generating series

$$a(t) = \sum_{i=0}^{\infty} \frac{F'(A_i)}{i!} t^{i+1},$$

this becomes

$$2t\,\partial_t \left(\frac{a}{t}\right) = \partial_t(a^2),$$

that is,

$$da - a\frac{dt}{t} = a\ da,$$

so

$$\frac{da}{a} - da = \frac{dt}{t}.$$

Since $a'(0) = 1$, we obtain

$$a\,e^{-a} = t.$$

Corollary (3.3).

$$F'(B_\ell) = \ell^\ell.$$

Here W is the subgroup of $\mathrm{GL}(\ell,\mathbb{R})$ preserving a set $\{\pm\varepsilon_i\}$. We have $|W| = 2^\ell \ell!$, $h = 2\ell$; for example,

$$c : \varepsilon_1 \mapsto \varepsilon_2 \mapsto \cdots \mapsto \varepsilon_\ell \mapsto -\varepsilon_1,$$

and the number of Coxeter elements is $2^{\ell-1}(\ell-1)!$.

For a product of ℓ reflections to be Coxeter, it is necessary and sufficient that:

(a) one and only one of them be of type ($\varepsilon_i \mapsto -\varepsilon_i$, ε_j fixed);

(b) the graph obtained by putting an edge for each reflection joining i and j by a reflection $\varepsilon_i \mapsto \pm\varepsilon_j$ be a tree.

Hence the number of sequences of ℓ reflections whose product is Coxeter is

$$\ell^{\ell-2}(\ell-1)!\,2^{\ell-1}\ell^2,$$

and the cardinality of each Ξ'_c is therefore

$$\frac{\ell^{\ell-2}(\ell-1)!\,2^{\ell-1}\ell^2}{2^{\ell-1}(\ell-1)!} = \ell^\ell.$$

3.4

Corollary 1.5 gives, with $B_0 = \varnothing$ and $B_1 = A_1$, for $\ell \geq 1$,

$$F'(B_\ell) = \ell \sum_{i+j=\ell-1} \frac{(\ell-1)!}{i!j!} F'(A_i)F'(B_j),$$

or

$$\frac{F'(B_\ell)}{\ell!} = \sum_{i+j=\ell-1} \frac{F'(A_i)}{i!} \frac{F'(B_j)}{j!}.$$

In terms of the generating series

$$b(t) = \sum_{i=0}^{\infty} \frac{F'(B_i)}{i!} t^i,$$

this becomes

$$b(t) = a(t)b(t) + 1,$$

that is,

$$b(t) = \frac{1}{1-a(t)}.$$

Corollary (3.5).

$$F'(D_\ell) = 2(\ell-1)^\ell \qquad (\ell \geq 2).$$

Let

$$d(t) = 1 + \sum_\ell \frac{F'(D_\ell)}{2\ell!} t^\ell$$

be the generating series. Corollary 1.5 gives

$$F'(D_\ell) = (\ell-1)\left[\sum_{\substack{i+j=\ell-1\\ j\geq 2}} \frac{(\ell-1)!}{i!j!} F'(A_i)F'(D_j) + 2F'(A_{\ell-1})\right],$$

hence

$$\ell \frac{F'(D_\ell)}{2\ell!} = (\ell-1)\left[\sum_{\substack{i+j=\ell-1\\ j\geq 2}} \frac{F'(A_i)}{i!}\frac{F'(D_j)}{2j!} + \frac{F'(A_{\ell-1})}{(\ell-1)!}\right].$$

Multiplying by $t^{\ell-1}$ and summing gives

$$\partial_t d = t\partial_t\left(\frac{ad}{t}\right),$$

or

$$\partial_t d = \partial_t d \cdot a + d \cdot t\partial_t\left(\frac{a}{t}\right), \qquad t\partial_t\left(\frac{a}{t}\right) = a\partial_t a.$$

Thus

$$\partial_t d(1-a) = d\, a\, \partial_t a,$$

$$\frac{\partial_t d}{d} = \frac{a\partial_t a}{1-a} = \partial_t\{-a - \log(1-a)\},$$

and

$$d = \frac{e^{-a}}{1-a}.$$

Since $ae^{-a} = t$, this may also be written as

$$d = \frac{t}{a(1-a)}, \qquad \frac{ad}{t} = b.$$

Substituting this in the differential equation gives

$$\partial_t d = t\partial_t b,$$

which proves the corollary.

I have also checked that

$$F'(E_6) = 2^9 3^4, \qquad F'(E_7) = 2 \cdot 3^{12}, \qquad F'(E_8) = 2 \cdot 3^5 \cdot 5^7.$$

Your conjecture is proved.

4. Complements

4.1

The following result sharpens 1.3(ii).

Let W be a finite Coxeter group with irreducible Dynkin diagram D. Let $\leq$ be a total order on D, and let α be a root. For each chamber C in which α is a simple root, let $\alpha_1, \ldots, \alpha_\ell$ be the simple roots associated with C, in the order $\leq$, and set

$$c(C) = z_\alpha z_{\alpha_1} \cdots \widehat{z_\alpha} \cdots z_{\alpha_\ell},$$

the product of the z_{α_i} in the order $\leq$, except that z_α is put first.

Proposition (4.2). *The map*

$$C \longmapsto c(C)$$

from the set of chambers in which α is a simple root to the set of Coxeter transformations is bijective.

Injectivity is proved as in 1.3(iv). With the notation of that proof, let $w \in W_d$ send C' to C''. It preserves α and the order of the walls, hence it preserves $c(C') = c(C'')$. Since w centralizes a Coxeter transformation and fixes a root, $w = 1$ by 1.3(i) and (v).

It remains to count. There are $|W|/h$ Coxeter transformations, by 1.3(i). If ℓ_α is the number of vertices of the Dynkin diagram corresponding to roots conjugate to α, then α has $\ell_\alpha h$ conjugates under W, by 1.3(v). The number of pairs

$$(\text{root } \beta \text{ conjugate to } \alpha, \ \text{chamber in which } \beta \text{ is a simple root})$$

is $|W|\ell_\alpha$; hence the number of chambers in which α is a simple root is $|W|/h$.

Yours,

P. Deligne